

Akshay Borse

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building and scaling large-scale platforms, distributed systems, and AI-powered products. Experience spans product development, platform engineering, architecture modernization, and applied AI, with systems operating at 100K+ TPS, platforms adopted across 60+ services, and products delivering \$28M+ in measured annualized impact.

EXPERIENCE

Senior Software Engineer | Microsoft | Redmond, WA

Sep 2025 to Aug 2026

- **Spearheaded Microsoft's flagship enterprise AI Agent Risk Detection platform**, orchestrating 90-day activity reconstruction across 40+ global environments to unlock instant day-one risk governance for 237,000+ AI agents across 13,000 enterprise customers.
- **Directed principal-level architecture across Purview IRM, Entra, and DLP** for Microsoft's flagship AI Adaptive Protection pipeline; authored the end-to-end technical spec and shipped core conditional access enforcement powering Microsoft 365 E7.
- **Pioneered hyper-scale distributed ingestion infrastructure with zero-latency fault isolation**, partitioning agent and human traffic across 28 worldwide deployments processing 150K+ critical security signals/day at 99.999% SLA.
- **Championed sovereign-cloud infrastructure expansion across US GCC, GCCH, and DoD high-security enclaves**, orchestrating 53 Azure resource groups and 51 deployments with automated disaster recovery and real-time Kusto telemetry.
- **Orchestrated unified investigation graphs across Entra, Defender, Sentinel, and Graph**, delivering unified threat intelligence that boosted customer security operation engagement by 28%.
- **Architected autonomous AI-agent synthetic validation and end-to-end test harnesses using Playwright**, simulating advanced attack vectors and policy violations across dynamic enterprise environments.
- **Pioneered closed-loop AI debugging and self-healing pipelines across 3 major codebases**, integrating telemetry pattern recognition with Playwright to automatically reproduce issues, generate code fixes, and submit PRs.

Software Engineer 2 | Amazon | Seattle, WA

Aug 2019 – Aug 2025

- **Architected enterprise 0→1 developer platform for 60+ Tier-1 microservices**, establishing standardized CDK-based infrastructure, zero-trust authentication, and CI/CD pipelines that reduced onboarding from 2 weeks to under 1 hour, saving 50+ engineering weeks.
- **Engineered monolith-to-microservices transformation at 100K+ peak TPS**, launching 15+ mission-critical microservices with high-performance gRPC orchestration handling 10M+ daily transactions at sub-10ms latency.
- **Directed ML-predicted delivery promise service powering global fulfillment**, integrating high-throughput quantile-regression models across planning and fulfillment systems delivering \$28M+ in annual bottom-line impact.
- **Transformed fulfillment risk operations with real-time ML decisioning on SageMaker**, deploying automated XGBoost risk scoring that prevented thousands of order cancellations and generated a 7% sales lift.
- **Architected real-time inventory intelligence platforms processing 200K+ events/day**, engineering Just-In-Stock and Restock alerting with distributed replay that increased restocked sales by 13% and expedited delivery 16%.
- **Pioneered production RAG-based product recommendation architecture for out-of-stock items**, combining embedding vector search with deterministic inventory constraints to deliver a 23% increase in conversion sales.
- **Spearheaded generative AI localization platform using AWS Bedrock**, implementing constrained vocabulary decoding and deterministic guardrails across 37 global notification templates across 28 locales with zero compliance errors.
- **Spearheaded operational excellence and resilience roadmap across 60+ distributed services**, mentoring 15+ engineers, cutting Sev2 incidents 40% in 60 days, and maintaining 100% uptime through Prime Day and Black Friday.
- **Architected synchronized fulfillment capabilities for high-volume enterprise Amazon Business orders**, eliminating manual workflows, freeing 300+ engineering hours/year, and boosting GCCP efficiency by 11%.
- **Championed container modernization and distributed observability across 60+ microservices**, designing shared container standards and optimized telemetry that reduced CloudWatch spend \$250K+/month and cut image size 83%.

Software Engineering Intern | Liquiron | San Jose, CA

Dec 2018 to Jan 2019

- **Migrated 2 backend services to serverless**, modernizing authentication with OAuth 2.0/Passport.js and Firebase.
- **Built the API layer for a single-page application**, delivering 4 REST APIs, 10 frontend capabilities and reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India

Sep 2016 to Jul 2017

- **Built a real-time CNN classification system for images and video**, achieving 87% accuracy across 7 sentiment categories.
- **Modernized legacy web applications into a single-page architecture**, improving response time 25% and unreliable-network performance 9%; introduced an engineering feedback system adopted by 6 teams.

Software Engineering Intern | Soft Corner | Pune, India

Jun 2015 to Jun 2016

- **Improved F2FS garbage-collection efficiency 15% through Linux kernel-level performance engineering**, instrumenting filesystem behavior and benchmarking cost-benefit and greedy cleaning policies to optimize latency.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. Sep 2017 to Jun 2019
- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. Jun 2012 to Jun 2016

FEATURED PROJECTS

- **LeetDesign**: Executable system-design simulator for modeling scalability, fault tolerance, workloads, and architectural trade-offs.
- **FinanceOS**: AI Financial Intelligence Platform: Designed a hybrid deterministic + LLM architecture that computes financial facts through deterministic pipelines and uses LLM reasoning only for explanation and recommendations, keeping probabilistic models outside correctness-critical calculations.

SKILLS

- **Languages**: Java • C# • Python • C++ • C • TypeScript • JavaScript • SQL
- **Architecture & Systems**: Distributed Systems • System Design • Microservices • Event-Driven Architecture • REST APIs • gRPC • Concurrency • Fault Tolerance • High Availability • Linux
- **Cloud & Platform**: AWS • Azure • Platform Engineering • Developer Platforms • Infrastructure as Code (IaC) • CI/CD • Docker • ECS • Lambda • Step Functions • DynamoDB • Cosmos DB • Observability • SLOs
- **AI/ML & Security**: Generative AI • LLMs • RAG • AI Agents • AI Infrastructure • AI Governance • Machine Learning • NLP • XGBoost • Bedrock • SageMaker • AI Security • DLP • Entra • Purview • Playwright